Product Vision and Visual Design System

Spec-set role: forensic reconstruction document generated from the three recovered HOTAS Control Panel chat notes. This document is meant to stand on its own while cross-referencing the rest of the set.

Evidence taxonomy used throughout:

? Confirmed Facts: directly stated in the recovery notes or visible as exact recovered UI text.
? Screenshot-Derived Details: observations recovered from screenshot descriptions in the source notes.
? Inferences: explicitly inferred from the notes or reconstructed from adjacent evidence.
? Desired Improvements: user preferences, future direction, fixes, or rebuild requirements.
? Unknowns: details the notes say are missing, uncertain, or need later verification.

Confirmed Facts

? The product target is premium HOTAS tuning and diagnostics software, not a rough mapper or toy utility.
? The modern V2 direction uses a dark professional desktop-control workspace with left sidebar, header/status area, assistant launcher, main workspace, scrollable pages, and footer.
? The desired reaction from the user is preserved exactly in the source notes: “DAMN THIS IS NICE SOFTWARE.”
? The product should communicate serious engineering/tuning value while remaining understandable.
? Rounded corners, smoothness, spacing discipline, professional copy, and status/action distinction are hard requirements.
? The final sidebar title is `HOTAS V2`; subtitle is `Control workspace`.
? Final header title is `HOTAS Control Panel V2`.
? Final header subtitle is `Professional tuning, live inspection, and assistant-guided diagnostics in one modern control workspace.`
? The flight-recorder chat also preserves a visible tagline: “Routing, tuning, live inspection, and guided diagnosis in one focused control workspace.”

Screenshot-Derived Details

? The final shell uses a left sidebar, top header, top-right `STATUS` cluster, top-right `ASSISTANT` cluster, scrollable main page area, and footer.
? Final top-right status chips include `Workspace Copy`, `Saved`, and `Idle` / `Live`.
? Final top-right assistant entry is `Helm`.
? Status chips must not look like action buttons.
? Action buttons include `Import Profile`, `Revert`, `Save Workspace`, and `Helm`.
? Final icon direction has two variants: a detailed installer/about-page icon and a simplified taskbar/Start Menu icon.

Inferences

? Exact font family was not recovered, but all sources point toward a modern sans-serif desktop UI with clear heading/body/caption hierarchy.
? Approximate palette uses deep navy/blue-black backgrounds, dark slate-blue cards, muted blue borders, cyan/blue action accents, green live/active states, amber caution states, and red destructive/disabled states.
? The product identity should be closer to engineering workstation / control console / signal lab than generic SaaS or game HUD.

Desired Improvements

? Avoid pixelated custom canvas-rounded corners.
? Use Qt/QSS/native styling for border radius and visual hierarchy.
? Avoid muddy dark strips behind every text area.
? Avoid debug/prototype wording such as `safe parallel Qt frontend`, `V2 draft`, and overly internal save-flow language in final user-facing copy.
? Keep animations subtle: page fade/slide, sidebar active transition, live/assistant pulse, Helm overlay transition.
? Avoid graph data animation and heavy effects that cause jank.

? Preserve visual seriousness: not bubbly, not toy-like, not generic SaaS.

Unknowns

? Exact font file.
? Exact finalized color-token values.
? Exact animation implementation.
? Exact final app-icon files and conversion path.
? Whether the simplified taskbar icon was ever converted into a multi-size `.ico`.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 1.2 Design philosophy`

1.2 Design philosophy

The project's visual and UX target changed dramatically during the chat. The desired direction became:

? Premium engineering / tuning software.
? Modern dark theme.
? Crisp, smooth, rounded, professional desktop feel.
? Real product identity, not a themed widget experiment.
? Users should open it and think: "DAMN THIS IS NICE SOFTWARE."

The user strongly preferred:

? Rounded corners throughout.
? Smooth animations and micro-interactions.
? High polish without jank.
? Clear visual hierarchy.
? Clear distinction between status indicators and action buttons.
? Polished product copy rather than debug/prototype wording.
? Real installer/app behavior rather than a desktop shortcut pointing into a messy support folder.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 2.20 App packaging / installation`

2.20 App packaging / installation

User goal: make the app feel like a real Windows app, not a desktop shortcut into a folder full of support files.

Recommended final deployment model:

? PySide6 app build.
? One-folder release build, not necessarily one-file.
? Inno Setup installer.
? Real `.ico` icon embedded in EXE/installer/shortcuts.
? Start Menu shortcut.
? Optional Desktop shortcut.
? Uninstaller entry.
? App files installed to a normal install directory:
  ? `%ProgramFiles%\HOTAS Control Panel V2`
  ? or `%LocalAppData%\Programs\HOTAS Control Panel V2`
? User data stored separately:
  ? `%AppData%\HOTASControlPanelV2`

? or `%LocalAppData%\HOTASControlPanelV2`

Important principle: app binaries/support files live in install folder; editable user data lives in AppData.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 2.21 App icon`

2.21 App icon

Two icon directions were generated/discussed:

1. Detailed icon
    - ? Dark rounded-square frame.
    - ? HOTAS throttle and joystick hardware.
    - ? Cyan/teal graph lines and tuning display.
    - ? Word `HOTAS` at bottom.
    - ? Premium, glossy, detailed.
    - ? Best for large installer/about-page/icon preview sizes.
    - ? Possible downside: too detailed at 16x16 or taskbar size.
2. Simplified/bolder taskbar icon
    - ? Dark navy rounded-square background.
    - ? Minimal white joystick silhouette.
    - ? Cyan gauge arc / response curve behind it.
    - ? Cyan bar/level indicators to the side.
    - ? No small detailed text.
    - ? Designed to read better at small taskbar/start-menu sizes.
    - ? Better for Windows taskbar and Start Menu.

Important note: Need convert simplified PNG into multi-size `.ico` with embedded sizes such as 16, 24, 32, 48, 64, 128, 256.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 3.1 General V2 window layout`

3.1 General V2 window layout

The final V2 UI uses a dark modern layout with:

- ? Left vertical sidebar.
- ? Main content workspace to the right.
- ? Top header across main content.
- ? Top-right status cluster and assistant launcher.
- ? Scrollable central page area.
- ? Bottom footer/action/status bar.
- ? Rounded cards/panels everywhere.
- ? Blue/cyan accent colors.
- ? Green live/active status colors.
- ? Red/destructive/caution states where needed.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 3.2 Color palette inferred`

3.2 Color palette inferred

Approximate colors/inferred roles:

? App background: very dark navy/blue-black.
? Sidebar/card surfaces: slightly lighter navy.
? Card border: muted blue/cyan stroke.
? Active nav background: richer blue.
? Active nav left accent: bright cyan vertical bar.
? Primary button: blue/cyan filled or bordered.
? Status live/active: green/teal glow.
? Warning/caution: amber/yellow/orange.
? Disabled/destructive: red/dark red.
? Text: white for headings, pale blue-gray for secondary text.
? Graph grid: subtle dark blue grid lines.
? Graph curves: cyan/blue, green, gray linear/reference.

Source-Preserved Block:
`hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` **/** `### 3.3 Typography`

3.3 Typography

No exact font file named. Inferred style:

? Modern sans-serif.
? Bold white headings.
? Smaller pale-blue helper/caption text.
? Monospace used in diagnostics areas and live values.
? Product copy became concise and polished.

Source-Preserved Block:
`hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` **/** `### 3.4 Rounded corners`

3.4 Rounded corners

Rounded corners became a hard preference:

? Buttons rounded.
? Cards rounded.
? Sidebar rounded.
? Header/status panels rounded.
? Inputs/dropdowns rounded.
? Overlay rounded.
? App icon rounded-square.

Important implementation rule: use Qt/QSS/native styling, not heavy custom canvas corner renderers. Avoid jagged/pixelated corners and expensive custom painting.

Source-Preserved Block:
`hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` **/** `### 3.5 Status vs button semantics`

3.5 Status vs button semantics

Problem noticed in early V2: status chips and action buttons looked too similar.

Final direction:

- ? Status chips should be small, quiet, non-button-like:
  - ? Workspace Copy
  - ? Saved
  - ? Idle/Live
- ? Action buttons should look clickable:
  - ? Import Profile
  - ? Revert
  - ? Save Workspace
  - ? Helm
- ? Primary actions should have stronger emphasis.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 3.6 Sidebar`

3.6 Sidebar

Final sidebar contents:

- ? Title: `HOTAS V2`
- ? Subtitle: `Control workspace`
- ? Navigation items:
  - ? Mapping
  - ? Modes
  - ? Base Tuning
  - ? Filtering
  - ? Combat Profile
  - ? Profiles
  - ? Conditional Rules
  - ? Effective Response Stack
  - ? Live Monitor
  - ? Help / Docs
  - ? Perf / Diagnostics
- ? Active page highlight:
  - ? Rounded blue item background.
  - ? Cyan border/outline.
  - ? Bright cyan vertical accent at left.
- ? Bottom live status card:
  - ? Dot/pulse indicator.
  - ? Label `Runtime`.
  - ? State `Idle` or `Live`.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 3.7 Header/top-right cluster`

3.7 Header/top-right cluster

Final header:

? Main title: `HOTAS Control Panel V2`

? Subtitle: `Professional tuning, live inspection, and assistant-guided diagnostics in one modern control workspace.`

? Top-right `STATUS` cluster:

  ? `Workspace Copy`

  ? `Saved`

  ? `Idle` or `Live`

? Top-right `ASSISTANT` cluster:

  ? `Helm` button.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 3.8 Footer`

3.8 Footer

Final footer includes:

? Left status message, e.g. "Reverted the workspace to the last saved or imported state."

? Center details:

  ? Page: mapping/base/filtering/combat/profiles/rules/stack/monitor/help.

  ? Axis: Roll.

  ? Profile: Current Workspace.

  ? Source: Workspace copy: `hotas_bridge_config_v2.json`.

? Right action buttons:

  ? `Import Profile`

  ? `Revert`

  ? `Save Workspace`

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 3.9 Animation/motion direction`

3.9 Animation/motion direction

Desired and partially implemented:

? Subtle page transition fade/fade+slide.

? Sidebar active-state transition.

? Live/assistant pulse/breathe indicator.

? Helm overlay open/close transition.

? Avoid graph data animation and heavy effects that cause jank.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### Screenshot 25 — Final V2 Mapping page, top overview`

Screenshot 25 — Final V2 Mapping page, top overview

Sidebar final text `HOTAS V2`, `Control workspace`. Mapping selected. Header final copy: "Professional tuning, live inspection, and assistant-guided diagnostics in one modern control workspace." Status cluster: STATUS Workspace Copy / Saved / Idle. Assistant cluster with Helm button. Mapping page shows routing overview, live route summary, axis routing table, footer actions.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### Screenshot 38 — Detailed app icon`

Screenshot 38 — Detailed app icon

Detailed rounded-square dark metallic icon with throttle, joystick, cyan response curve, graph display, and `HOTAS` text. Premium but too detailed for tiny taskbar sizes.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### Screenshot 39 — Simplified taskbar app icon`

Screenshot 39 — Simplified taskbar app icon

Simpler dark navy rounded-square icon with white joystick silhouette, cyan gauge/curve arc, and cyan bar indicators. Best for taskbar/Start Menu scale. Needs conversion to multi-size `.ico`.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 10.10 V2 early: dark strips behind text looked awful`

10.10 V2 early: dark strips behind text looked awful

? Symptom: every text area had darker strip background; muddy/striped.
? Fix: simplify surface hierarchy: app background, card background, control/input background.
? Status: final screenshots much cleaner.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 10.11 V2 copy too debug/prototype-like`

10.11 V2 copy too debug/prototype-like

? Symptom: text like “safe parallel Qt frontend,” “V2 draft,” “this page writes only...” sounded dev/test.
? Fix: product-grade copy pass.
? Status: final copy is much more polished, though some status terms like Workspace Copy remain intentionally visible.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 11.1 Things the user liked`

11.1 Things the user liked

? PySide6/Qt V2 direction.
? Modern dark premium UI.
? Rounded corners everywhere.
? New Mapping page.
? New Profiles page.
? Live Monitor page.
? Help/Docs page after improvement.

? Effective Response Stack as a premium/crown-jewel page.
? Helm as overlay assistant rather than sidebar tab.
? Pulsing green live/Helm indicator.
? Graph integration with pyqtgraph.
? Clear active sidebar highlight.
? Status/footer live indicators.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` **/** `### 11.2 Things the user disliked`

11.2 Things the user disliked

? Legacy UI looking like "hardcoded decent html at best."
? Pixelated/jagged rounded corners.
? Slow tab switching.
? Redraw skipping/loading effect.
? Missing scrollbars/unreachable content.
? Overlap/clipping.
? Bottom action bar taking too much room.
? Darker strips behind all text.
? Status chips looking like buttons.
? Helm suggestions looking like action buttons.
? Machine/copy-paste Helm output.
? Debug/test/internal wording.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` **/** `### 11.3 Performance expectations`

11.3 Performance expectations

? Smooth tab switching.
? No obvious redraw jank.
? No visible whole-page repaint on scroll/tab switch.
? Graph updates scoped to graph.
? Hidden pages do not do expensive work.
? Performance instrumentation visible.
? Animations tasteful and not expensive.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` **/** `### 11.4 Packaging preferences`

11.4 Packaging preferences

? Real app installation.
? User sees one app/shortcut, not folder full of files.
? Real app icon.
? Start Menu/desktop shortcuts.
? AppData for user configs/profiles/logs.
? Installer/uninstaller.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` **/** `### Design philosophy inferred from the chat`

Design philosophy inferred from the chat

The design philosophy was strongly shaped by the following preferences and decisions:

? The app should feel like serious software, not a rough hobby utility.
? The UI should be premium, modern, dark, polished, and intentional.
? Advanced functionality should exist, but it should not explode across the main interface.
? The user repeatedly wanted the current pages to stay beautiful, clean, and preserved.
? Main workflows should remain simple, while deeper controls should be hidden behind things like Configure dialogs.
? Features should be useful, not gimmicky.
? New systems should feel integrated into the product instead of bolted on.
? The app should support both live monitoring and after-the-fact analysis.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` **/** `### Key goals and UX priorities explicitly visible in the chat`

Key goals and UX priorities explicitly visible in the chat

? Preserve the existing Flight Recorder page and avoid wrecking its layout.
? Add a real live overlay without cluttering the recorder page.
? Put advanced overlay controls behind a Configure entry point.
? Use sane defaults so the user rarely needs to adjust overlay specifics.
? Reuse shared overlay logic across replay/export and live rendering.
? Keep the app looking polished and organized.
? Support a hindsight / buffered replay style capture mode.
? Keep visual style consistent between replay overlays and live telemetry overlays.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` **/** `### Overall visual style`

Overall visual style

The app has a strong, premium dark theme with the following characteristics:

? very dark navy/blue background,
? slightly lighter blue-gray cards,
? bright white headings,
? muted blue-gray descriptive text,
? neon-ish accent chips in cyan/green/orange/yellow tones,
? subtle borders,
? rounded corners throughout,
? layered card-based layout,
? clean spacing,
? a modern desktop-tool feel rather than a game-like HUD aesthetic.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` **/** `### Visual language`

Visual language

The UI communicates “control workspace” rather than “toy utility.” It uses:

? large rounded cards,
? pill-shaped badges and inputs,
? slim bright progress bars,
? sharp but soft-edged panels,
? consistent margin spacing,
? and a restrained, high-contrast palette.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` **/** `### Typography`

Typography

Direct font name is not stated.

Inference: likely a modern sans-serif desktop UI font. Text styling suggests hierarchy such as:
? large bold page header,
? medium bold section titles,
? lighter gray body/supporting text,
? uppercase small labels for areas like STATUS and ASSISTANT.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` **/** `### Colors observed`

Colors observed

Approximate visual palette:
? Background: deep navy / blue-black
? Cards: dark slate-blue
? Borders: muted blue lines
? Accent blue: used for buttons, outlines, progress bars, and selection states
? Green: used for active/ready/live states
? Amber/orange: used for unsaved or alert-like status chips
? Axis colors:
  ? Roll: light blue (#58B8FF visible)
  ? Pitch: mint/green (#6FDB9F visible)
  ? Throttle: warm yellow/orange (#F0C46A visible)
  ? Yaw: lavender/violet (#CF95FF visible)
  ? Aux 1: orange/coral (#FF9B6B visible)
  ? Aux 2: aqua/cyan-green (#6ED9D0 visible)

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` **/** `### Component style`

Component style

? Buttons are often rounded rectangular or pill-like.
? Inputs are dark fill with lighter outline.
? Cards have large radius corners.
? Panels use thin borders rather than strong shadows.
? The overall tone is polished and technical.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` / `### Specific UX preferences explicitly stated in the chat`

Specific UX preferences explicitly stated in the chat

? Do not overload Flight Recorder with lots of overlay placement controls.
? Keep current pages intact if they already look good.
? Add advanced settings only behind Configure.
? Main controls should remain compact and obvious.
? Sane defaults matter.
? The interface should feel premium and intentional.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_flight_recorder_live_overlay_and_helm.md` / `### Before vs after changes discussed`

Before vs after changes discussed

Before

? Flight Recorder existed with on-demand recording, overlay export, library, preview, and axis overlay options.
? The user worried that adding a live overlay might require a lot of layout changes or too many settings.

Design decision phase

? Assistant proposed a separate live overlay path launched from Live Monitor, backed by a shared overlay engine, with advanced settings hidden behind Configure.

After

? Live Overlay was successfully implemented.
? Flight Recorder remained intact.
? Live Monitor gained a Live Overlay card.
? A full configuration dialog exists.
? Flight Recorder gained buffered replay / save previous interval mode.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### Design philosophy`

Design philosophy

The app's design philosophy evolved into something very specific:

? technical, enthusiast-grade, control-room aesthetic
? dark UI with blue-accent styling
? highly visual and inspectable

- ? powerful, but still understandable
- ? not toy-like, not generic SaaS, not childish
- ? problem-solving oriented, not just parameter-editing oriented
- ? built for people who care about feel, latency, and control precision

The project intentionally became more than “a settings page.” It was discussed as a serious diagnostic/tuning workstation. Several times the design target was described as a premium control console, signal lab, avionics/control-room tool, or engineering workstation rather than a generic utility.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### User experience priorities`

User experience priorities

Explicit priorities repeatedly mentioned:

- ? input tuning must help with difficult real-game problems like:
  - ? “can’t hold aim steady”
  - ? “too twitchy near center”
  - ? “combat feels sluggish”
  - ? “overshoots target”
- ? UI must look and feel premium
- ? UI must move smoothly and not look like pages are manually redrawn
- ? CPU and GPU usage should be kept very low
- ? system latency should be minimized aggressively
- ? visuals should be modernized without turning the app into a bubbly dashboard
- ? components should be highly inspectable and richly visualized
- ? there should be a clear distinction between:
  - ? parameter-driven tuning tabs
  - ? problem-driven smart diagnosis (Helm)

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### Key constraints and preferences`

Key constraints and preferences

- ? Preserve technical feel and “serious tool” identity
- ? Avoid overly playful or cartoonish UI
- ? Avoid fluffy, generic consumer-SaaS styling
- ? Avoid over-reliance on static presets when giving smart recommendations
- ? Keep control processing responsive even while rich visualization is active
- ? Rich UI must never compromise low-latency control behavior
- ? Use modular architecture for advanced features like Helm rather than stuffing logic into a monolithic app file

---

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### 2.10 Visual Modernization Project`

2.10 Visual Modernization Project

Why it exists

Even after features and performance improved, the app still felt visually like “2005 manually drawn and redrawn graphics.” User wanted:

? more modern visuals
? smoother movement
? subtle curve on sharp edges
? soft shadows
? less flat boxy legacy feeling
? more premium control-console identity

Core modernization goals discussed

? slight but visible rounded corners
? soft shadows
? strong surface hierarchy
? better spacing and fit
? stronger panel/card language
? modernized tabs, buttons, inputs, headers
? not bubbly, not SaaS, not toy-like

Additional explicit requirement

Spacing and layout discipline were called out later as first-class needs:

? no overlap
? consistent spacing rhythm
? no cramped fit
? no clipping
? cards/graphs/Helm sections must breathe properly

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### Colors and style`

Colors and style

? dark background
? blue accent color family
? user repeatedly praised the dark theme and felt it was clean and technical
? later wanted stronger surface hierarchy, rounded corners, shadows, blur, and premium depth
? wanted the whole app to stay technical and control-console-like

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### Borders, corners, shadows`

Borders, corners, shadows

Originally boxy / square / legacy-framed. Later strong request to modernize with:

? slightly rounded corners everywhere
? stronger but tasteful shadows
? parent/child surface distinction
? more premium selected/active states
? more depth on Helm and graph panels

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` **/** `### Typography`

Typography

No exact fonts specified. Desired text style:

- ? technical, clean, premium
- ? more modern hierarchy
- ? stronger spacing between headings/subtitles/body
- ? less flat text rhythm

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` **/** `### Icons`

Icons

Mentioned/described:

- ? top-right Helm icon/button
- ? drawer/docs/help icon later requested for help drawer
- ? rule/status chips and indicators
- ? possible helm/nautical identity for assistant

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` **/** `### Tooltips`

Tooltips

Strongly preferred for explaining advanced fields. The small tooltip icon approach on Conditional Rules was later requested across the app.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` **/** `### Visual hierarchy desires`

Visual hierarchy desires

- ? panels should feel like true surfaces
- ? graph panels should feel like instruments
- ? stack cards should feel like modules
- ? Helm should feel like a different operating mode

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` **/** `### Bug/problem: visual modernization too subtle`

Bug/problem: visual modernization too subtle

Symptom

After Phase 2, app did not look different enough.

Fix requested

A much bolder Phase 2.5:

? stronger radius

? stronger shadows

? stronger panel hierarchy

? better spacing and fit

? more premium shell/header

? better graph/stack/Helm surface treatment

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### Explicit likes`

Explicit likes

? dark technical UI

? blue accents

? graph previews and live visualizations

? grouped sections and plain-English previews

? Effective Response Stack as flagship feature

? Helm as a named character/workspace, not generic assistant

? top-right Helm entry point

? large slide-out workspace mode for Helm

? control-console / premium workstation feel

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### Explicit dislikes`

Explicit dislikes

? overly boxy / legacy 2005 appearance

? cold third-person wording in Helm

? suggestions that feel too preset or timid

? vague context notes

? tiny cramped helper overlays

? side help text instead of small tooltips

? visual chunking / page redraw feeling

? overlapping / cramped / barely-fitting layout

? generic SaaS or bubbly styling

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### Emotional/practical priorities`

Emotional/practical priorities

? project was almost complete before being lost; reconstruction needs to be dense and forensic

? performance and low latency matter deeply

? user wants humor in normal conversation but app tone itself should remain serious/professional with character

? wants rich explainability and trust in what the app is doing

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / `### Visual style preferences`

Visual style preferences

? modernized but still sharp
? slightly rounded, not pill-shaped
? shadows and depth, but disciplined
? strong spacing discipline
? polished motion later
? control-room identity preserved

Second Sweep Addendum: Legacy Shell, Support Surfaces, and Visual Coverage

Confirmed Facts

The second source comparison found several small visual/support observations that were not lost functionally, but were not preserved as exact wording in the first spec pass. They are retained here because they help rebuild the original-to-V2 visual evolution and the support/diagnostics surfaces.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / `### 2.1 Legacy HOTAS Control Panel`

2.1 Legacy HOTAS Control Panel

What it was: the original/legacy UI before V2. It used a dark sci-fi control-console style with cyan borders, tabbed sections, and many nested custom frames. It contained the core tuning features and was nearly complete before the user's original files were erased.

Primary tabs/features visible in legacy screenshots:

? Modes
? Base Tuning
? Filtering
? Combat Profile
? Conditional Rules
? Effective Response Stack
? Helm
? Help / Docs
? Performance popup

Known files changed during legacy refactors:

? `hotas_control_panel_app.py`
? `app_performance.py`
? `hotas_ui_theme.py`
? `Open-HOTAS-Control-Panel.cmd`
? `Edit-HOTAS-Curves.cmd`
? Packaged output: `HOTASControlPanel.exe`

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_rebuild_and_productization.md` / missing support screenshots

Screenshot 8 — Legacy Performance popup

Top-right `Perf On`. Popup titled `Performance`, close button. Text: "Lightweight timings for the live UI paths. Keep it off during normal tuning." Checkbox `Collect live timings`, button Clear. Current live diagnostics show Visible tab Stack, Selected axis Roll, runtime timestamp unavailable, pending refresh counts, next scheduler tick.

Screenshot 9 — Legacy Help / Tuning Notes popup

Popup on right side titled `Help & Tuning Notes`, close button. Cards explain `Curves` and `Deadzone / Anti-Deadzone / Hysteresis`. Old help panel is narrow and has excessive dead space.

Screenshot 23 — Initial V2 Help / Docs page

Help content was simple text page with product philosophy and page overview. Later upgraded into searchable topic tree.

Screenshot 24 — Initial V2 Perf / Diagnostics page

Diagnostics page with dense timing text: active page, runtime live, hidden page skips, page build/switch timings, heartbeat stats, graph timings. Confirms performance instrumentation was built in.

Screenshot 36 — Final V2 Help / Docs page

Help / Docs selected. Search bar, Sort by Category dropdown, Topics tree, Guide pane. Conditional Rules article selected. Looks like integrated product documentation.

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / shell layout notes

Overall window layout

The app is a dark desktop application with a strong top shell/header and multiple main tabs. The bottom has an action bar with things like save/reload/close/perf/help depending on state. The central area changes depending on tab/workspace.

Core top-level areas observed/discussed

? top shell/header with app title and status/info area
? main tab strip for tuning/inspection tabs
? content body
? right-side graph/diagnostic panels on many pages
? bottom action bar
? top-right special controls including Helm, perf, maybe help/docs

Known major tabs/pages

? Modes
? Base Tuning
? Filtering
? Combat Profile
? Conditional Rules
? Effective Response Stack
? Helm is *not* a normal tab; later it becomes a special overlay workspace

Source-Preserved Block: `hotas_control_panel_recovery_notes_chat_v_2_hotas_control_panel_helm_recovery.md` / screenshot 7

Screenshot 7 — Phase 2 visual modernization comparison (descriptive placeholder)

? User reaction: only slight visual changes, not strong enough. Still too close to original design.

? Resulting decision: create a bolder Phase 2.5 visual modernization emphasizing stronger radius, shadows, spacing, fit, and hierarchy.

> Note: Actual screenshot embedding was not available in Canvas. Reattach screenshots manually later to these placeholders.